

Visualization of Bias of Machine Learning for Content Recommendation

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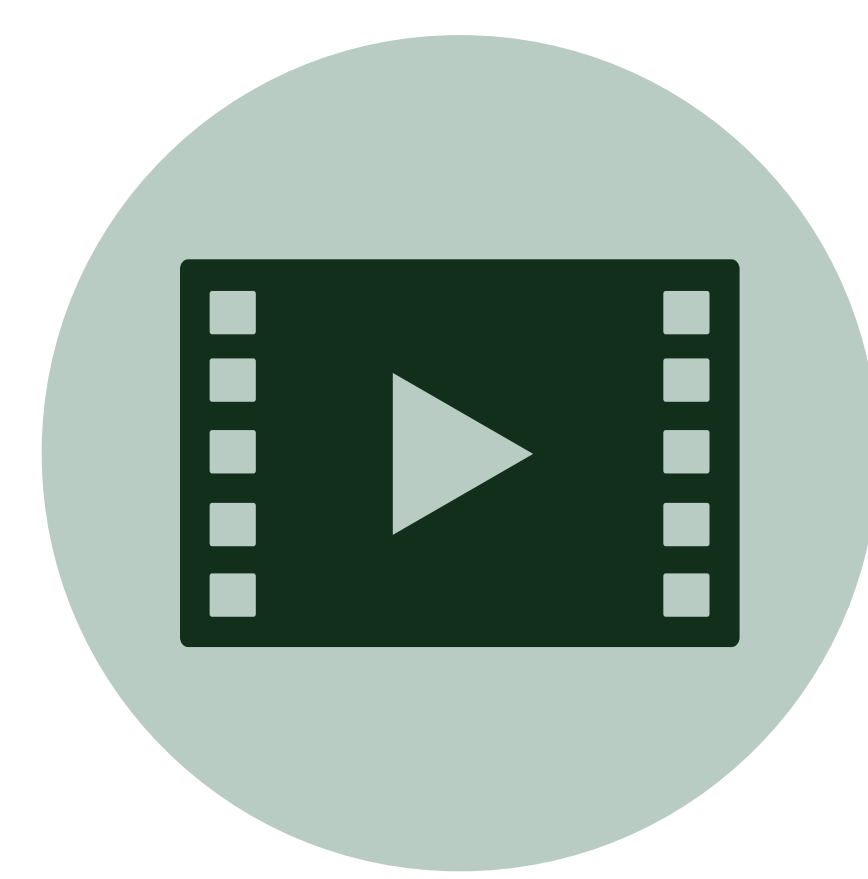
Introduction

- ✓ Excessively personalized recommendation systems may cause biased recommendation results
- ✓ Machine learning techniques have been applied to recommendation engines in recent years

This study

Propose a visualization system to assist the comparison between statistics of appreciation and recommendation of the particular contents/customers and the discovery of the biases in learning results.

VISUALIZATION SYSTEM



Dataset
MovieLens 1M

USERS 6040

MOVIES 3883

BPR
BPR+GAN

Learning Models

test dataset

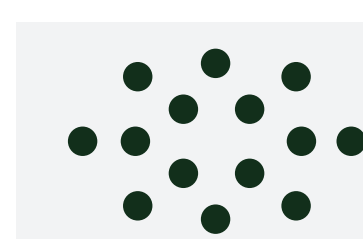
× 20

RECOMMEND!

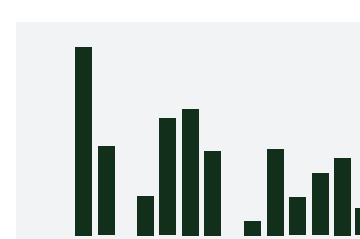


customer

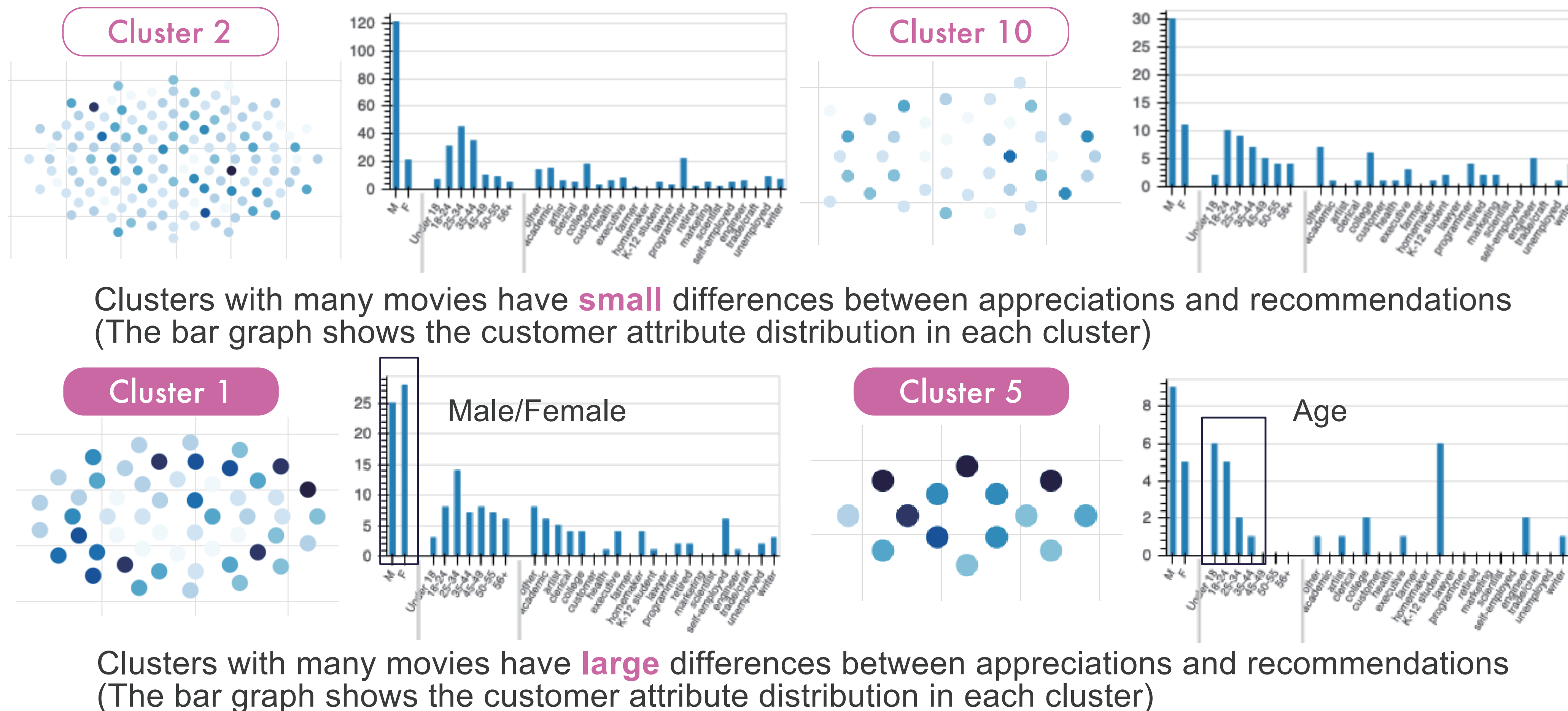
CASE STUDY



VIS01 SCATTERPLOT
Selected particular cluster



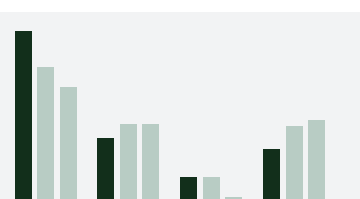
VIS02 CLUSTER BAR
Customer distribution in the selected cluster



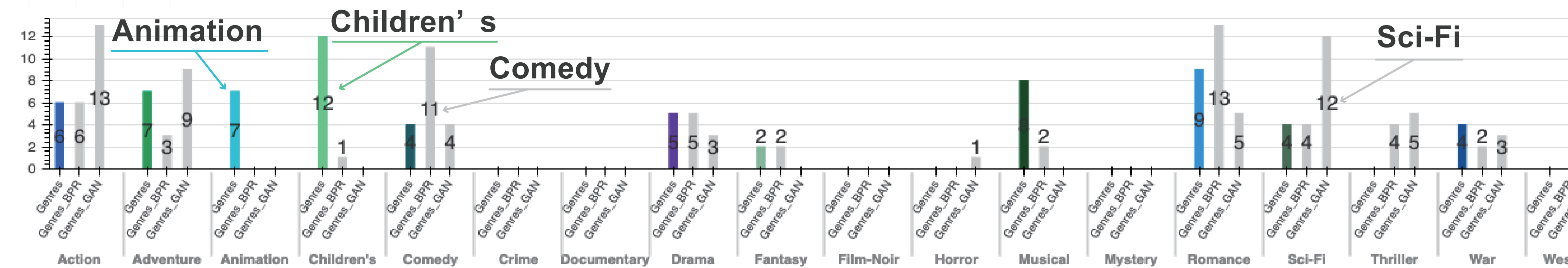
F, 18-24,
college/grad student



VIS01 SCATTERPLOT
Selected particular node



VIS03 ATTRIBUTES BAR
Movie distribution watched by customer nodes



Conclusion

This study presented a visualization tool aiming at the discovery of bias of machine learning results of the recommendation systems. Results of case study suggests that the difference in recommendation results among the clusters is caused by the differences in attributes of customers.

Future Work

- ✓ Improvement of calculation method of similarity between nodes
- ✓ Comparison between learning models
- ✓ Selection of quantitative evaluation values